Reg. No: | | | | | | | | | | | |

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccreditedby NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

COMMON TO B.E – CSE & M.TECH - CSE

20CS211 - ALGORITHMS AND COMPLEXITY - II

Answer ALL questions

Duration: 3 Hours **Maximum: 100 Marks**

PART - A **(7x 1 = 7 Marks)**

1. **CO1** ________ is a measure of the maximum amount of time an algorithm takes to execute a specific set of inputs.

   a) Best Case Time Complexity b) Worst Case Time Complexity

   c) Average Case Time Complexity d) Maximum Time Complexity

2. **CO2** Which of the following is a self-adjusting or self-balancing Binary Search Tree ?

   a) Splay Tree b) AVL Tree c) Red Black Tree d) All of the above

3. **CO2** Which of the following is the most desirable condition for interpolation search?

   a) array should be sorted

   b) array should not be sorted but the values should be uniformly distributed

   c) array should have a less than 64 elements

   d) array should be sorted and the values should be uniformly distributed

4. **CO3** Which of the following algorithms can be used to most efficiently determine the presence of a
   cycle in a given graph?

   a) Depth First Search b) Breadth First Search

   c) Prim's Minimum Spanning Tree Algorithm

   d) Kruskal's Minimum Spanning Tree Algorithm

5. **CO4** When a top-down approach of dynamic programming is applied to a problem, it usually ________.

a) Decreases both, the time complexity and the space complexity

b) Decreases the time complexity and increases the space complexity

c) Increases the time complexity and decreases the space complexity

d) Increases both, the time complexity and the space complexity

6. **CO3** Choose the correct statement from the following.

a) branch and bound is not suitable where a greedy algorithm is not applicable

b) branch and bound divides a problem into at least 2 new restricted sub problems

c) backtracking divides a problem into at least 2 new restricted sub problems

d) branch and bound is more efficient than backtracking

7. **CO4** Dijkstra's Algorithm is the prime example for ________.

a) Dynamic programming b) Back tracking

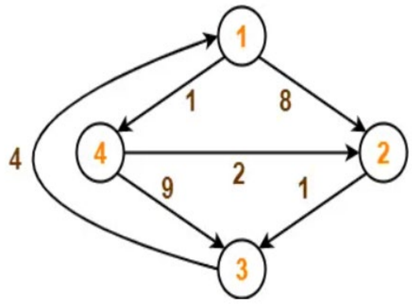
c) Branch and bound d) Greedy algorithm

Fill in the blanks: **(3x 1 = 3 Marks)**

8. **CO2** ________ happens when the backtracking algorithm reaches a complete solution.
9. **CO3** The worst-case efficiency of solving a problem in polynomial time is _________.
10. **CO5** Problems that can be solved in polynomial time are known as _________.

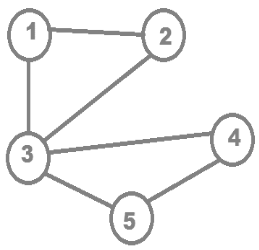
PART - B **(5x 3 = 15 Marks)**

11. **CO1** Illustrate the process of inserting into an AVL Tree using the following data: 12,8,5,4,11,18,17
12. **CO2** One of the 99 identically looking coins is fake. The fake coin differs in weight from the original ones, but it is not known whether the fake coin is lighter or heavier than the rest. How can you determine, in two weightings, if the fake coin is lighter or heavier? What if you had 101 coins?
13. **CO3** Consider the below given directed weighted graph. Using Floyd Warshall's Algorithm, find the shortest path distance between every pair of vertices.



14. **CO4** Illustrate the solution for 4-Queens problem, using backtracking. The problem is to place the 4 queens in a 4x4 chessboard, with no 2 queens attacking each other.

15. **CO5** What is a clique in the context of graphs? For the graph given below, find out all the possible cliques, and also identify the maximal clique of the graph.



PART - C **(5 x 15 = 75 Marks)**

Q.No: 16 Compulsory Question

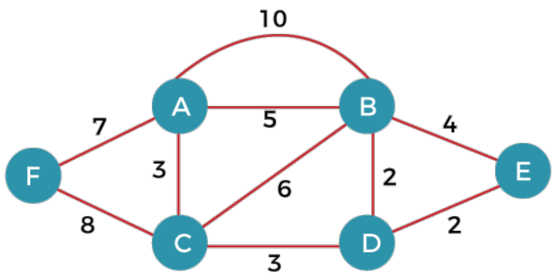
Q.No: 17 to 22 Answer any <u>FOUR</u> Questions

<u>Compulsory Question:</u>

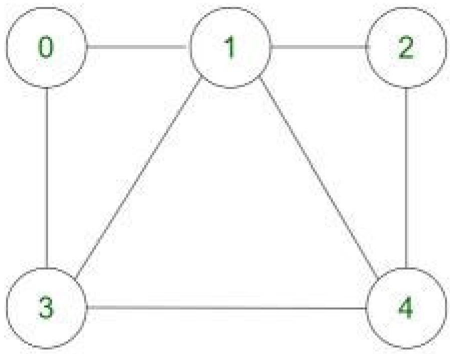
16. **CO1** With the elements given below: 41,16,51,13,100,31. Build a Min-Heap and Using Heap Sort technique, sort the elements in ascending order. (15)

17. **CO2** Give a detailed overview of how insertion sort works over the given number sequence to generate a sorted sequence in ascending order.: 56,41,92,84,13,35,14,52,20,18 (15)

18. **CO3** What is the importance of Kruskal's algorithm? Illustrate the operations of Kruskal's algorithm for the weighted graph given below. (15)



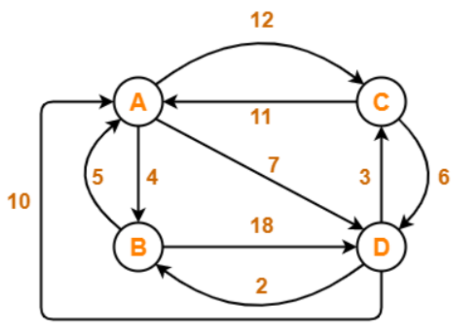
19. **CO4** For the graph given below: (15)



Define the terms Hamiltonian Path and Hamiltonian Circuit. Why is finding a Hamiltonian cycle in a graph said to be as an NP complete Problem? Identify the Hamiltonian Path and Circuit for the graph shown above.

20. **CO4** i) Differentiate between P, NP, NP Complete problems (5)

CO3 ii) Rahul is a tour consultant. He is assigned to figure out an optimal plan to cover some major cities of Central India, for a team of designated foreign visitors. The graph given below shows various routes between the cities and the distance in kilometers. Using proper techniques, help Rahul to identify the best routes to cover all the cities with minimum cost incurred. (10)



21. **CO5** What is the time complexity of the fractional knapsack problem solved using greedy method? Substantiate your answer based on the following scenario. Given the weights W and values V of N items which are to be inserted in the knapsack which can bear the weight upto K. Insert maximum value in the knapsack with weight not greater than K. Every item can also be inserted in fractional manner (i.e, we can break the item to maximize the value in the knapsack). (15)

Input: arr= {(20,4), (63,7), (40,8), (30,5)} and K= 18

22. **CO5** Discuss in detail about Binary Search Trees and Optimal Binary Search Trees, focusing on their applications, strengths and limitations. (15)

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